

CHARGE SOL

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ABMCA22194

For the award of the degree of
Master Of Computer Applications

Under the supervision of
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Session: 2024

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1. Introduction

1.1 Overview

The primary goal is to establish charging stations that can accommodate any electric vehicle (EV), regardless of its brand. This approach is crucial for accelerating the transition from internal combustion engine (ICE) vehicles to EVs, addressing the numerous environmental, economic, and public health challenges posed by ICE vehicles. These challenges include high greenhouse gas emissions, air pollution, reliance on non-renewable fossil fuels, and resulting economic instability.

Objectives for Universal EV Charging Stations

1. Compatibility and Standardization

- a. **Universal Access:** Ensure that all EVs can charge at any station, regardless of make or model, by adopting standardized connectors and communication protocols.
- b. **Interoperability:** Implement industry-wide standards to facilitate seamless interaction between different EV models and charging infrastructure.

2. Environmental Sustainability

- a. **Emission Reduction:** Promote EV use to decrease greenhouse gas emissions and improve air quality, thus combating climate change and protecting public health.
- b. **Renewable Energy Integration:** Encourage the use of renewable energy sources for EV charging to further reduce the transportation sector's carbon footprint.

3. Energy Efficiency

- a. **Optimized Energy Use:** Improve the efficiency of energy conversion from the grid to vehicle movement, reducing overall energy consumption compared to ICE vehicles.

- b. Smart Charging Solutions: Develop smart charging technologies to optimize energy use, manage grid load, and lower charging costs.

4. Economic Advantages

- a. Lower Operating Costs: Reduce maintenance and fuel costs for consumers, making EV ownership more economically attractive.
- b. Energy Independence: Reduce reliance on imported oil, enhancing national energy security and economic stability.
- c. Innovation and Growth: Stimulate technological innovation in EV-related fields, promoting economic growth and securing a strong position in the global market.

5. Public Health Benefits

- a. Reduced Pollution: Lower air pollution levels, leading to fewer respiratory and cardiovascular health issues.
- b. Healthcare Savings: Reduce healthcare costs by decreasing pollution-related health problems.

6. Supportive Policies and Incentives

- a. Government Support: Utilize financial incentives, tax credits, and grants to encourage EV adoption and infrastructure development.
- b. Regulatory Framework: Advocate for stricter emissions regulations to create a favorable market environment for EV growth.

7. Consumer Demand and Market Dynamics

- a. Increased Awareness: Promote awareness about the benefits of EVs and sustainable living to drive consumer demand.
- b. Infrastructure Development: Expand and improve charging infrastructure to make EVs a more practical and convenient choice for consumers.

1.2 Objective:

1. ChargeSol, is a groundbreaking project aimed at revolutionizing India's electric vehicle (EV) ecosystem. With a vision to become the largest community and marketplace for the EV industry, ChargeSol focuses on enabling seamless roaming and interoperability among EV owners and charging station operators. By providing a comprehensive platform, ChargeSol aims to facilitate easy access to charging stations, thus accelerating the transition towards sustainable transportation solutions.
2. ChargeSol's mission is to empower EV owners with a seamless charging experience, thereby fostering the widespread adoption of electric vehicles across India. With a vision to create a network of 100,000 charging stations accessible to 5 million EV owners, ChargeSol aims to play a pivotal role in shaping a sustainable future for transportation. By providing innovative solutions and fostering collaboration among key stakeholders, ChargeSol endeavors to lead the charge towards a greener and cleaner tomorrow.
3. With aim to provide an integrated platform that builds a vast network of aggregated chargers to provide access to numerous eMobility partners. Our unified open roaming network aims to bridge the gap between Charge point operators and Vehicle manufacturers to provide a seamless charging experience that knows no geographical bounds and complicated negotiations.

1.3 Mission

Our mission is to enable 5 million EV owners to travel hassle free with a charging network of 100,000 chargers on a single app around the country by providing a seamless charging experience across the networks of multiple CPOs

1.4 Functions

- Energy Analytics.
- Power of Integration.
- Payment Processing.
- Multi-level User Management.
- Control & Monitor Remotely.

2. Requirement Specification

This section highlights the functional requirements, non-functional requirements and other requirements.

2.1 Functional Requirements

2.1.1 Performance requirements

- User Satisfaction: The system is such that it stands up to the user expectations.
- Response Time: The response of all operations is good.
- Error Handling: Response to user errors and undesired situation has been taken care of to ensure that the system operates without halting.

2.1.2 Compatibility and Interoperability

- Standardized Connectors: Charging stations must support universal connectors compatible with all EV models.
- Communication Protocols: Implement industry-standard communication protocols (e.g., OCPP - Open Charge Point Protocol) to ensure seamless interaction between different EVs and the charging infrastructure.

2.1.3 Charging Station Network

- Scalability: The system should be scalable to accommodate increasing numbers of EVs and charging stations.
- Network Integration: Charging stations must be integrated into a network for centralized management, monitoring, and control.

2.1.4 Charging Options

- **Multiple Charging Levels:** Provide various charging levels (e.g., Level 1, Level 2, and DC fast charging) to accommodate different needs and EV capabilities.
- **Adaptive Charging:** Implement smart charging features that can adjust the charging rate based on grid demand, EV battery status, and user preferences.

2.1.5 User Interface and Experience

- **User-Friendly Interface:** Charging stations must have an intuitive and accessible user interface, both on the physical unit and via mobile apps.
- **Mobile App Integration:** Offer a mobile application that allows users to locate charging stations, check availability, start/stop charging, and monitor charging progress.

2.1.6 Payment and Billing

- **Multiple Payment Options:** Support various payment methods, including credit/debit cards, mobile payments, and subscription services.
- **Transparent Billing:** Provide clear and detailed billing information, including the cost of charging and any additional fees.

2.1.7 Energy Management

- **Renewable Energy Integration:** Enable the use of renewable energy sources for charging where possible.
- **Load Balancing:** Implement load balancing to distribute electricity demand evenly across the network and prevent grid overloads.
- **Energy Storage:** Integrate energy storage solutions (e.g., batteries) to store excess renewable energy and use it during peak demand periods.

2.1.8 Security and Data Privacy

- **Secure Communication:** Ensure that all data transmitted between the EV, charging station, and network is encrypted and secure.
- **User Data Privacy:** Protect user data in compliance with relevant privacy regulations and policies.

2.1.9 Maintenance and Support

- **Remote Diagnostics:** Enable remote diagnostics and troubleshooting to quickly address any issues with the charging stations.
- **Regular Updates:** Provide regular software updates to enhance functionality, security, and compatibility

2.1.10 Environmental and Safety Standards

- **Safety Compliance:** Ensure all charging stations meet safety standards and regulations, including protection against electrical hazards.
- **Environmental Compliance:** Design stations to minimize environmental impact, including the use of sustainable materials and energy-efficient components.

2.1.11 Reporting and Analytics

- **Usage Reports:** Generate detailed reports on charging station usage, including frequency, duration, and energy consumption.
- **Performance Analytics:** Monitor and analyze performance metrics to optimize station operations and user satisfaction.

2.1.12 Accessibility

- **Physical Accessibility:** Design charging stations to be accessible to users with disabilities, following relevant accessibility standards.
- **Language Support:** Provide multi-language support in both the user interface and mobile applications.

2.2 Non-Functional Requirements

- **Hardware Requirements:**
 - Internet connection
 - Android 5 or above
 - Ram 2gb or above
- **Software Requirements:**
 - Chrome or any compatible browser
 - Internet
 - Application

2.3 Feasibility study

Technical Feasibility

Availability:

- Objective: Ensure uninterrupted access to the eRoaming network for EV drivers and CPOs.
- Strategy: Implement redundant server architecture and cloud-based infrastructure to minimize downtime. Utilize load balancing and failover mechanisms to handle traffic spikes and server failures.

Security:

- Objective: Safeguard user data, transaction information, and network integrity against cyber threats and unauthorized access.
- Strategy: Employ robust encryption protocols, multi-factor authentication, and intrusion detection systems. Conduct regular security audits and penetration testing to identify and mitigate vulnerabilities.

Reliability:

- Objective: Deliver consistent and dependable charging services to EV drivers, ensuring high satisfaction and trust.
- Strategy: Establish Service Level Agreements (SLAs) with CPOs to guarantee uptime and performance standards. Implement real-time monitoring and predictive maintenance to preemptively address potential system failures.

Portability:

- Objective: Enable seamless access to the eRoaming network across various geographic locations and charging infrastructures.
- Strategy: Develop a user-friendly mobile application compatible with iOS and Android devices. Ensure interoperability with different EV models and charging station interfaces through standardized protocols.

Maintainability:

- Objective: Facilitate easy maintenance and updates to the eRoaming platform to ensure long-term sustainability and scalability.
- Strategy: Adopt modular design principles and microservices architecture to facilitate independent development and deployment of system components. Implement automated testing and continuous integration pipelines to streamline software updates and bug fixes.

Operational Feasibility

Implementation Plan:

- Objective: Roll out the eRoaming network in a phased approach, ensuring smooth integration and adoption.
- Strategy: Initiate pilot testing in select regions to gather feedback and fine-tune the platform. Gradually expand coverage to additional areas based on user demand and market readiness.

Resource Allocation:

- Objective: Allocate resources effectively to support platform development, marketing, and customer support.

- Strategy: Invest in skilled IT and engineering teams for platform development and maintenance. Allocate budget for marketing campaigns to attract users and partners. Establish a dedicated customer support team to address user queries and technical issues.

Support and Maintenance:

- Objective: Provide ongoing support and maintenance to ensure optimal performance of the eRoaming platform.
- Strategy: Implement a 24/7 customer support system to handle user queries and technical issues. Establish regular maintenance schedules to update software and address any system vulnerabilities.

Financial Feasibility

Cost Analysis:

- Objective: Assess the financial viability of developing and operating the eRoaming network.
- Strategy: Conduct a comprehensive cost analysis, including initial development costs, operational expenses, and potential revenue streams from user subscriptions and transaction fees.

Revenue Streams:

- Objective: Identify potential revenue streams to sustain the operation of the eRoaming network.
- Strategy: Explore revenue opportunities such as subscription fees from CPOs, transaction fees from charging sessions, and data analytics services for partners.

Break-even Analysis:

- Objective: Determine the timeline for achieving profitability and recouping initial investment.
- Strategy: Conduct a break-even analysis to estimate the time it will take to cover costs and generate positive cash flow based on projected user adoption and revenue generation.

Conclusion:

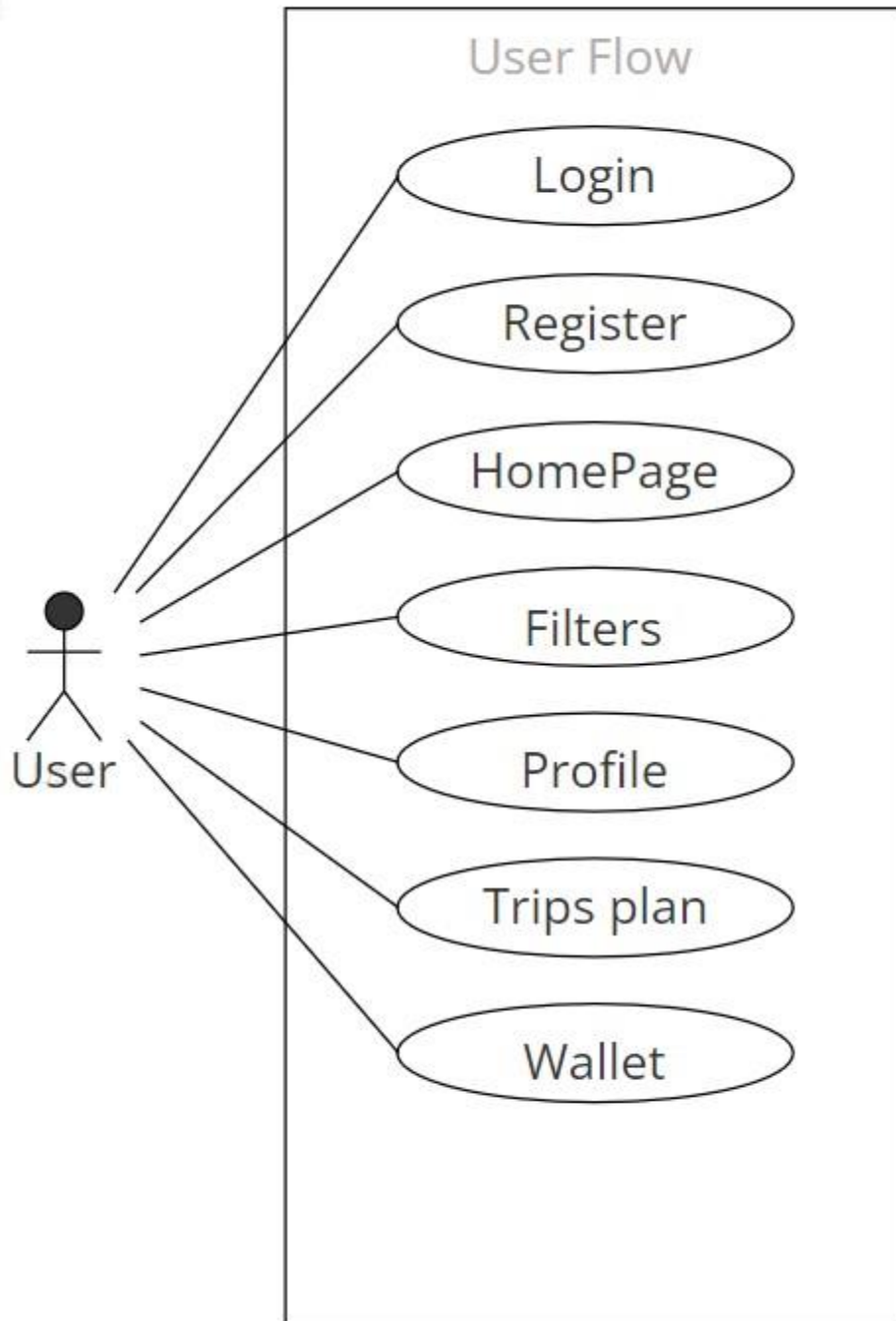
ChargeSol's mission to empower EV owners with a seamless charging experience aligns with its vision to create a network of 100,000 charging stations accessible to 5 million EV owners, shaping a sustainable future for transportation in India. By providing innovative solutions and fostering collaboration among key stakeholders, ChargeSol aims to lead the charge towards a greener and cleaner tomorrow. Through its integrated platform and unified open roaming network, ChargeSol bridges the gap between Charge Point Operators and Vehicle manufacturers, providing a seamless charging experience that transcends geographical boundaries and complex negotiations. With a steadfast commitment to technical feasibility, operational feasibility, and financial feasibility, ChargeSol is poised to revolutionize the eMobility market and accelerate the widespread adoption of electric vehicles.

2.4 Product Function

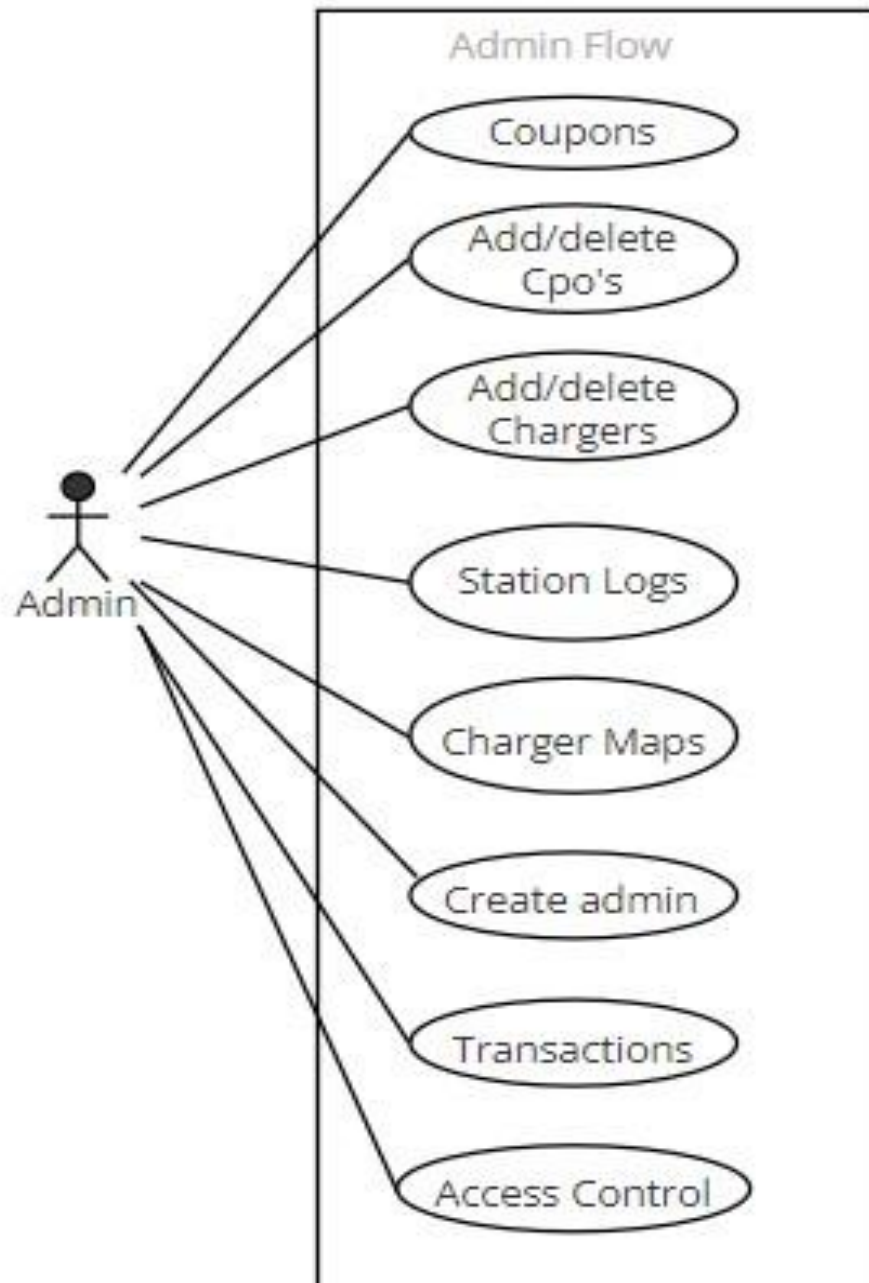
ChargeSol's eRoaming network provides a unified charging platform that integrates multiple Charge Point Operators (CPOs), simplifying the charging process for EV drivers by eliminating the need for multiple apps and accounts. The platform displays real-time status of charging stations, allowing drivers to locate and access available chargers easily, thereby reducing wait times and improving convenience. Seamless payment processing through the app ensures a hassle-free payment experience with real-time reconciliation and immediate crediting of payments to CPOs. Interoperability with various EV models and charging station types through standardized protocols enables drivers to use any charging station within the network, regardless of the vehicle or charger brand. Data-driven insights offer analytics on charging patterns, station usage, and transaction data, helping CPOs and stakeholders make informed decisions on infrastructure development and operational improvements. The network also provides 24/7 customer support to address technical issues and queries, ensuring a reliable support system for users. Additionally, ChargeSol equips CPOs and partners with tools to effectively expand the charging network, making it more accessible. Roaming capabilities connect with international eRoaming networks, allowing drivers to charge their vehicles seamlessly across different regions and countries. By focusing on these functions, ChargeSol aims to enhance the EV charging experience, promote EV adoption, and support sustainable transportation growth.

2.5 Use-case diagrams

User



1. Admin



3. System Design

3.1. High-Level Design

1.1 System Architecture

The system architecture of ChargeSol encompasses the key components and their interactions necessary to achieve the project's objectives.

1.1.1 Architecture Diagram

Provide a high-level architecture diagram that outlines the major components of the system and their interactions. This diagram should include components such as the central management system, charging stations, mobile applications, and external integrations.

1.1.2 Component Descriptions

Central Management System:

- **Functionality:** Manages overall system operations, including user management, charging station monitoring, and data analytics.
- **Interfaces:** Interfaces with charging stations, mobile applications, and external systems for data exchange and control.

Charging Stations:

- **Functionality:** Provide charging services to EVs, communicate with the central management system, and support various charging levels.
- **Interfaces:** Connect to the central management system and EVs using standardized protocols.

Mobile Application:

- **Functionality:** Allows users to locate charging stations, initiate and monitor charging sessions, and process payments.
- **Interfaces:** Communicates with the central management system and charging stations.

External Integrations:

- **Functionality:** Integrates with external systems such as payment gateways, renewable energy sources, and government databases.
- **Interfaces:** Use APIs and standardized communication protocols for seamless integration.

1.2 Modules and Subsystems

The system is divided into several modules and subsystems, each responsible for specific functionalities.

1.2.1 Module Overview

User Management Module:

- **Functionality:** Manages user accounts, authentication, and authorization.
- **Interfaces:** Interfaces with the mobile application and central management system.

Charging Management Module:

- **Functionality:** Manages charging sessions, monitors station status, and handles data communication.
- **Interfaces:** Interfaces with charging stations and the central management system.

3.3 Database Design

```
import mongoose from 'mongoose';
import crypto from 'crypto';
import bcrypt from 'bcryptjs';
import validator from 'validator';

const userSchema = new mongoose.Schema({
  name: {
    type: String,
    required: [true, 'A user must have a name'],
    maxlength: [20, 'Username must be less than or equal to 10 characters.'],
  },
  email: {
    type: String,
    required: [true, 'Please provide your email'],
    unique: true,
    validate: [validator.isEmail, 'Please provide a valid email'],
  },
  password: {
    type: String,
    required: [true, 'Please provide a password'],
    minlength: 8
  },
  notification: {
    type: String,
    default: 1
  }
});
```

```
import mongoose from 'mongoose';
const userSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true,
    unique: true,
  },
  permissions: {
    Dashboard: {
      type: Boolean,
      default: false,
    },
    Overview: {
      type: Boolean,
      default: false,
    },
    Cpos: {
      type: Boolean,
      default: false,
    },
    Chargers: {
      type: Boolean,
      default: false,
    },
  },
});
```

```
import mongoose from 'mongoose'; Akshaypareek01, 11 months ago • working on use
import crypto from 'crypto';
import bcrypt from 'bcryptjs';
import validator from 'validator';

const userSchema = new mongoose.Schema({
  name: {
    type: String,
    required: [true, 'A user must have a name'],
    maxlength: [20, 'Username must be less than or equal to 10 characters.'],
  },
  email: {
    type: String,
    required: [true, 'Please provide your email'],
    unique: true,
    validate: [validator.isEmail, 'Please provide a valid email']
  },
  password: {
    type: String,
    required: [true, 'Please provide a password'],
    minlength: 8
  },
  notification: {
    type: String,
    default: 1
  }
});
```

```
import mongoose from 'mongoose'; Akshaypareek01, 10 months ago
💡

const chargingStationSchema = new mongoose.Schema({
  cpoId: {
    type: mongoose.Schema.Types.ObjectId,
    ref: 'CpoUsers',
    required: true,
  },
  StationName: {
    type: String,
    required: true
  },
  Latitude: {
    type: String,
    required: true
  },
  Longitude: {
    type: String,
    required: true
  },
  street: {
    type: String,
    required: true
  },
});
```

```
import mongoose from 'mongoose';  
const msgSchema = new mongoose.Schema({  
  message: {  
    type: Object  
  },  
  sender_id: {  
    type: Object,  
    required: true  
  },  
  receiver_id: {  
    type: Object,  
    required: true  
  },  
  file_upload: {  
    type: Object  
  },  
  unread: {  
    type: Object,  
    default: '0'  
  },  
  flag: {  
    type: Object,  
    default: '0'  
  }  
}, {  
  timestamps: {  
    createdAt: true,  
    updatedAt: true  
  }  
});
```

```
const groupmsgSchema = new mongoose.Schema({  
  message: {  
    type: Object  
  },  
  sender_id: {  
    type: Object,  
    required: true  
  },  
  group_id: {  
    type: Object,  
    required: true  
  },  
  file_upload: {  
    type: Object  
  },  
  unread: {  
    type: Object,  
    default: 0  
  }  
}, {  
  timestamps: {  
    createdAt: true,  
    updatedAt: true  
  }  
});
```

```
const groupSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true
  },
  description: {
    type: String,
    required: true
  },
  userId: {
    type: String,
    required: true
  },
});

export const Groups = mongoose.model('groups', groupSchema);
```

```
add a breakpoint
const CustomerWalletSchema = new mongoose.Schema({
  customerId: {
    type: mongoose.Schema.Types.ObjectId,
    ref: 'CustomerModel', // Assuming your Customer model is named 'Customer'
    required: true,
  },
  balance: {
    type: Number,
    default: 0,
  },
  // Additional fields if needed
});

const CustomerWalletModel = mongoose.model('CustomerWallets', CustomerWalletSchema);

export default CustomerWalletModel;
```

```

name: {
  type: String,
  required: [true, "A user must have a name"],
  maxlength: [20, "Username must be less than or equal to 10 characters."],
},
email: {
  type: String,
  required: [true, "Please provide your email"],
  unique: true,
  validate: [validator.isEmail, "Please provide a valid email"],
},
password: {
  type: String,
  required: [true, "Please provide a password"],
  minlength: 8,
},
Brand_Name: { type: String, required: true },
GST_No: { type: String, required: true },
MID: { type: String, required: true },
Registered_Address: { type: String, required: true },
state: { type: String, required: true },
regional: { type: Boolean,
  default: false },

```

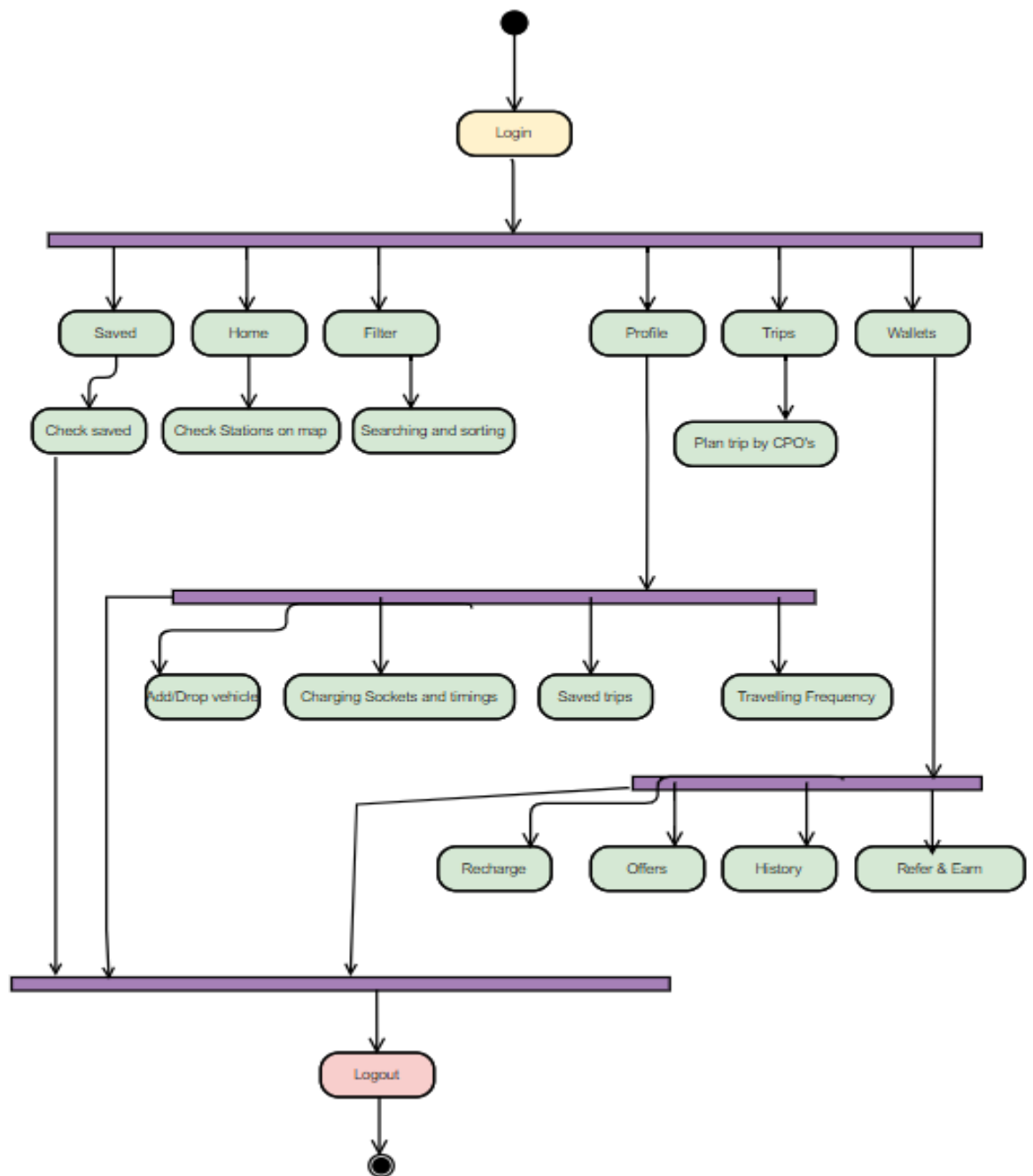
```

import mongoose from 'mongoose';
import crypto from 'crypto';
import bcrypt from 'bcryptjs';
import validator from 'validator';

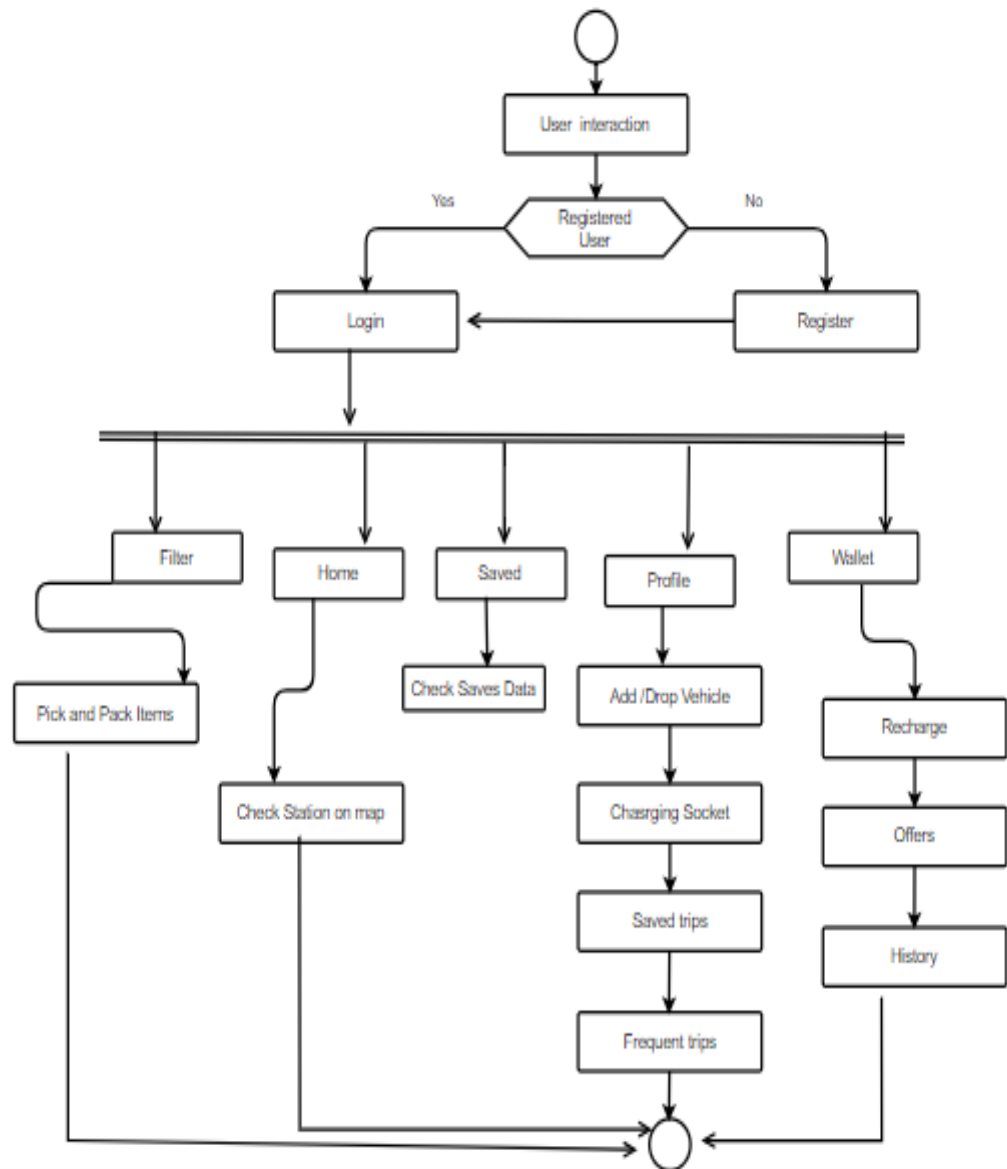
const contactSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true
  },
  email: {
    type: String,
    required: [true, 'Please provide your email'],
    validate: [validator.isEmail, 'Please provide a valid email']
  },
  user_id: {
    type: String,
    require: true
  },
  created_by: {
    type: String,
    require: true
  },
  last_msg_date: Date
});

```


3.4 Activity Diagram



3.5 Flow Chart



4. User Interface



Login

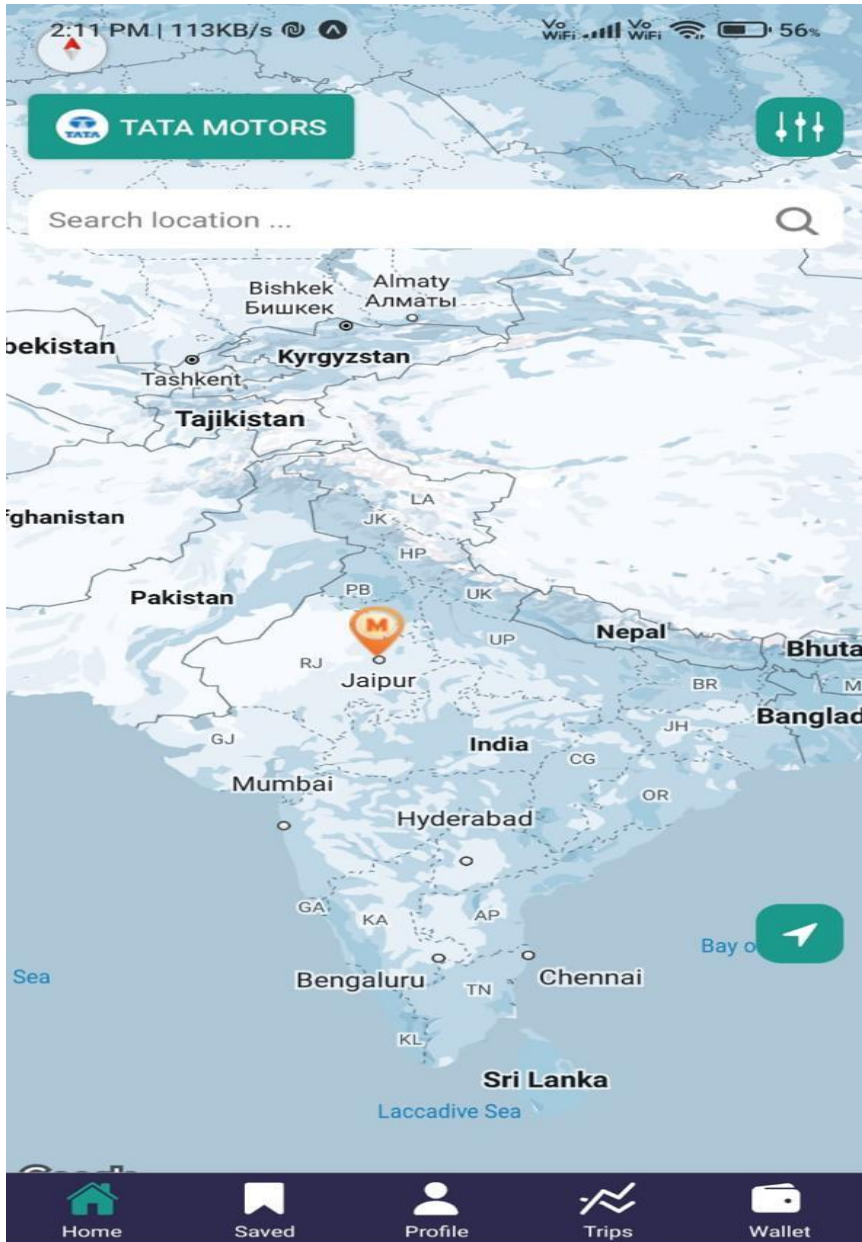
SKIP 



+91 Phone Number

Enter Pin

Don't have an account?
Sign up



 SignUp



+91 Phone Number

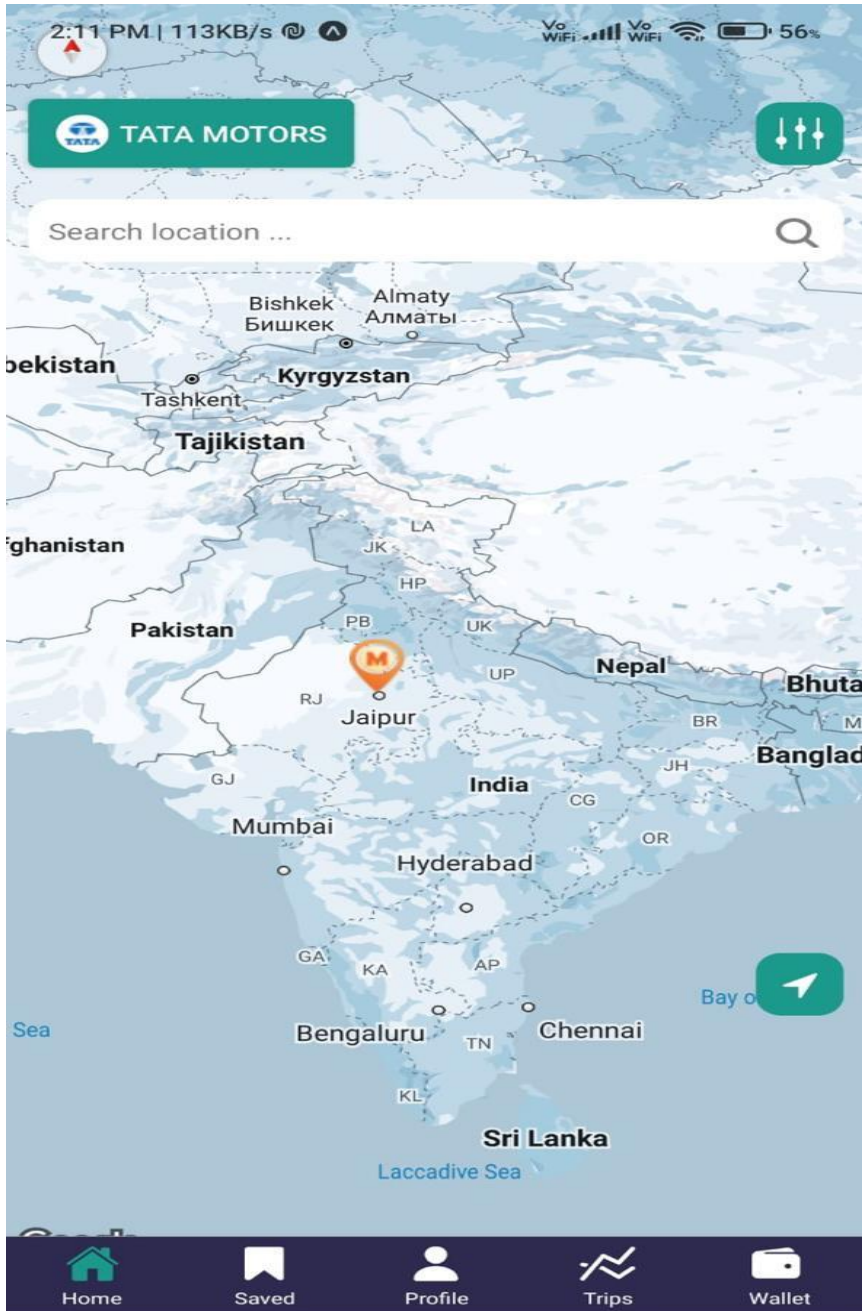
Full Name

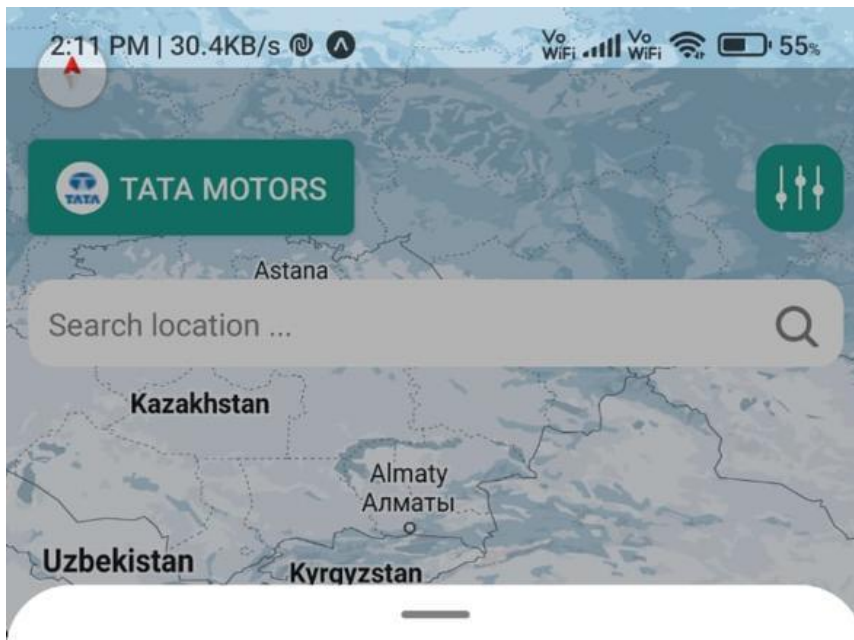
Password

Confirm Password

By tapping on 'Sign Up' , you agree to the Terms of Use and Privacy Policy

Sign Up





Filter

Clear All

Open Hours

Open now

Open 24/7

Autocharge Enabled



Amenities

Restroom

Cafe

Store

Ratings

Rating 4.0+



Rating 3.5+



Rating 3.0+



Apply

← Charger Detail



 **Disconnected**

Connector 1
(Type-2)

 **Surya Nagar**
Gopal Pura Mode,Jaipur,Rajasthan

 Copy Location

 Get Direction



Rest Room



Restaurant

[View More](#)

Customer Reviews

[see all](#)

4.7 out of 5

10 reviews



CHARGE NOW

← Charger Detail



U Unavailable



TATA Power

AC | 32kW

₹ 10/unit



Available

Connector 1
(CCS-2)

AC | 7.4kW

₹ 8/kwh



Disconnected

Connector 1
(Type-2)

Saved

Test 1

★ 3.5 (22 Reviews)

Surya Nagar, Gopal Pura Mode

42 min (5 km)

U Unavailable



Home



Saved



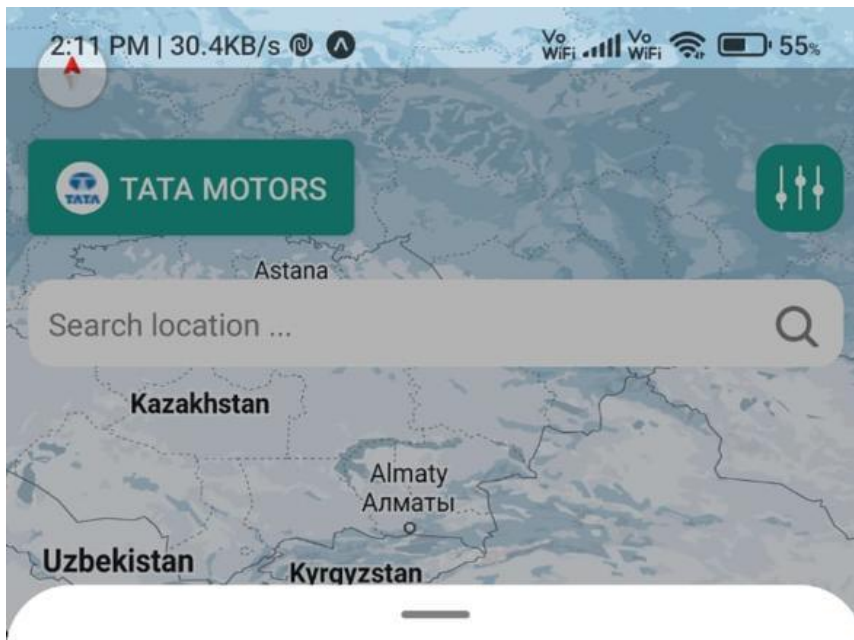
Profile



Trips



Wallet



Filter

Clear All

My Vehicles



Connectors



Type-2



Wall



CCS-2

Status

Active

Inactive

Charger Type

Apply

Wallet



Hi, Akshay !

welcome,back

Wallet

₹ 12,000

Phone Number

+91 8290918154

Member Since

03/23

Functions



RECHARGE



OFFERS



HISTORY

Refer and earn

₹300 cashback

Cashback will be credited in your app wallet



ICXXXX



Recent transactions



Home



Saved



Profile



Trips



Wallet

Trips



Home



Saved



Profile

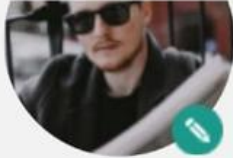


Trips



Wallet

Profile



@ test@gmail.com

8290918154



57

Total Km



187

kg Co Saved



14

Balance

My Vehicles



Tata Tiago EV

Mileage range 315 km



MG Z5 EV

Mileage range 315 km

Tata Xpres-T EV

Mileage range 295 km

Delete



Add Vehicle



Home



Saved



Profile

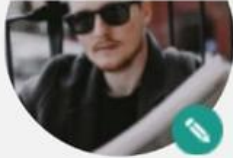


Trips



Wallet

Profile



@ test@gmail.com

8290918154



57

Total Km



187

kg Co Saved



14

Balance

My Vehicles



Tata Tiago EV

Mileage range 315 km



MG Z5 EV

Mileage range 315 km



Tata Xpres-T EV

Mileage range 295 km



Add Vehicle



Home



Saved



Profile

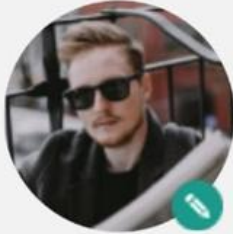


Trips



Wallet

Profile



Akshay



@ test@gmail.com

8290918154



57

Total Km



187

kg Co Saved



14

Balance

My Vehicles



Tata Tiago EV

Mileage range 315 km



MG Z5 EV

Mileage range 315 km



Add Vehicle



Charging



Saved Trips



Frequent



Home



Saved



Profile



Trips



Wallet

← Charger Detail



Plug In and Press Start

Time

Units

Percentage

How much time do you want to charge ?



 Apply Coupon

[View Offers](#)

Estimated Price

₹ 290.00

charge point CP123 is not
connected

Charge Now

← Charger Detail



Plug In and Press Start

Time

Units

Percentage

How much time do you want to charge ?



 Apply Coupon

[View Offers](#)

Estimated Price

₹ 290.00

Charge Now

← Charger Detail



Plug In and Press Start

Time

Units

Percentage

How much time do you want to charge ?



 Apply Coupon

[View Offers](#)

Estimated Price

₹ 290.00

Charge Now

← Charger Detail



Plug In and Press Start

Time

Units

Percentage

How much time do you want to charge ?



 Apply Coupon

[View Offers](#)

Estimated Price

₹ 290.00

Charge Now

← Charger Detail



Plug In and Press Start

Time

Units

Percentage

How much time do you want to charge ?



0

0 min

 Apply Coupon

[View Offers](#)

Estimated Price

₹ 0.00

Charge Now

← Charger Detail



 **Disconnected**

Connector 1
(Type-2)

📍 **Surya Nagar**
Gopal Pura Mode,Jaipur,Rajasthan

 Copy Location

 Get Direction



Rest Room



Restaurant

[View More](#)

Customer Reviews

[see all](#)

4.7 out of 5

10 reviews



CHARGE NOW

← Charger Detail



U Unavailable



TATA Power

AC | 32kW

₹ 10/unit



Available

Connector 1
(CCS-2)

AC | 7.4kW

₹ 8/kwh

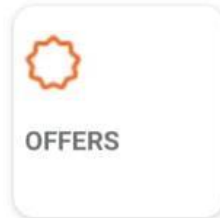


Disconnected

Connector 1
(Type-2)

Wallet

Functions



Refer and earn ₹300 cashback

Cashback will be credited in your app wallet



ICXXXX 

Recent transactions

Wallet Top up Master Card	+ ₹ 1,000 09/09/2023
Wallet Top up Master Card	+ ₹ 1,000 09/09/2023
Wallet Top up Master Card	+ ₹ 1,000 09/09/2023
Wallet Top up Master Card	+ ₹ 1,000 09/09/2023



Home



Saved



Profile



Trips



Wallet

Wallet

Wallet
₹ 12,000

Phone Number
+91 8290918154

Member Since
03/23

Functions

Credits ₹

1 credit = 1 INR

Amount

Enter Amount

+300

+500

+1000

Total Price

₹ 0

Proceed to Pay

← Offers



No coupons available

Wallet



Hi, Akshay !

welcome,back

Wallet

₹ 12,000

Phone Number

+91 8290918154

Member Since

03/23

Functions



Share

PERSONAL

WORK



WhatsApp



Chat



Gmail



Notes



Cancel

5. Appendices

ChargeSol's eRoaming network aims to enhance the EV charging experience, promote EV adoption, and support sustainable transportation through its unified platform, real-time availability, seamless payments, interoperability, data-driven insights, user support, network expansion tools, and roaming capabilities.

6.References

<https://chargesol.in/demo/>

[Open charge point protocol - Open Charge Alliance](#)